Fiscal Year:	FY 2017	**Task Last Updated:**	FY 05/17/2017
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Individual Differences in the Neurochemical and Behavioral Response to Exposure to Protons and HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	10/23/2012	**End Date:**	04/22/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 4/22/2017 per S. Monk/LaRC (Ed., 9/29/16) NOTE: Extended to 10/22/2016 per PI and NSSC information (Ed., 8/28/15)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Shukitt-Hale, Barbara Ph.D. (US Department of Agriculture/Agricultural Research Service, Beltsville)		
Grant/Contract No.:	NNX13AB73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neural and behavioral functioning after the conclusion of the mission. The objectives of the proposed research program are: (1) to evaluate the effects of repeated subthreshold exposures to HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative stress, inflammation, signaling molecules, and autophagy in key brain regions; and (3) to determine how individual characteristics, including age and sex, may influence the responsiveness of the individual to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our data indicate that there is a significant risk that exposure to low, NASA-relevant doses of HZE particles produces alterations in central nervous system (CNS) function and in the behaviors that are dependent upon the integrity of specific CNS systems. However, current research suggests several caveats that might modulate the effectiveness of such exposures in disrupting neuronal function and cognitive performance. First, the 56Fe particle-induced oxidative stress affects consolidation and not acquisition; that is, it disrupts the transfer of material from short-term to long-term memory and does not affect the ability of the organism to learn new material. Further, the disruption of consolidation can be prevented by prior administration of anti-oxidant diets containing blueberry extract. This diet prevents the disruption of both the behavior and the disruption of the underlying neuronal function. Second, the total dose of HZE particles and protons to which astronauts will be exposed on exploratory class missions will be delivered over the extent of the mission, which can be 2.5 to 3 years. It is unlikely that the dose of radiation experienced at any single time would be sufficient to affect cognitive performance. The present experiments continue and complete a series of pilot experiments designed to provide the proper procedures to analyze the effects of repeated sub-threshold doses of HZE particles on cognitive performance. The results obtained to-date indicate that the effects of repeated subthreshold exposures may be task-specific. However, these experiments were intended as pilot experiments to determine the procedures needed to provide an accurate evaluation of the effectiveness of repeated subthreshold exposures to HZE particles. The third experiment was a continuation of previous research to establish the threshold for the disruption of cognitive performance and CNS function following exposure to 4He particles. While the neurochemical analyses have not been completed yet, behavioral testing indicates that cognitive performance can be disrupted by doses as low as 0.025 cGy. The low dose needed to disrupt cognitive performance may reflect the fact that a greater number of particles would be needed to achieve this dose (fluence) than would be needed for higher LET (linear energy transfer) particles. Nonetheless, this may be worrisome because the majority of the dose to which astronauts within the spacecraft will be exposed will be comprised of low LET 4He and 1H particles, As such, there is the possibility that exposure to low doses of these particles will affect cognitive performance. The final series of experiments, also ongoing, continues our evaluation of the interaction between age and the development of performance decrements following exposure to HZE particles. Previous research using higher LET 56Fe and 48Ti indicated that older subjects were more sensitive to exposure to HZE particles (i.e., showed a performance decrement at a lower dose). Exposure to lower LET 16O particles, which had a lower threshold for the disruption of cognitive performance in young organisms, did not show a similar relationship between age and the dose needed to disrupt cognitive performance. As a result, it was hypothesized that exposure to low LET 4He particles would also not show a similar age-dependent relationship between age of the subject and the dose needed to produce a deficit in performance. Preliminary data for 4He particles are consistent with that hypothesis. Overall, the pattern of responding suggests that older organisms are more sensitive to exposure to high LET HZE particles: exposure to lower doses of high LET HZE particles in older organisms initiates a cascade of changes in CNS function similar to those observed in aged organisms and which ultimately affecting cognitive performance. A similar pattern may not follow exposure to low LET particles such as16O and 4He. As such the present results suggest that age of irradiation may be a confounding factor in attempting to evaluate the long-term degenerative consequences of exposure to HZE particles and must be considered as a part of risk analysis for changes in cognitive performance during exploratory class missions, particularly for exposure to higher LET particles. NOTE May 2017: Our ongoing experiments are continuing in the new grant, ‘Individual Differences in the Neuronal and Behavioral Response to Exposure to HZE Particles,’ grant NNX16AE06G. See that project for future reporting.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Peer-reviewed Journals	Poulose SM, Rabin BM, Bielinski DF, Kelly ME, Miller MG, Thanthaeng N, Shukitt-Hale B. "Neurochemical differences in learning and memory paradigms among rats supplemented with anthocyanin-rich blueberry diets and exposed to acute doses of (56)Fe particles." Life Sci Space Res (Amst). 2017 Feb;12:16-23. Epub 2016 Dec 15. http://dx.doi.org/10.1016/j.lssr.2016.12.002 ; PubMed PMID: 28212704 , Feb-2017
Articles in Peer-reviewed Journals	Rabin BM, Carrihill-Knoll KL, Miller MG, Shukitt-Hale B. "Age as a factor in the responsiveness of the organism to the disruption of cognitive performance by exposure to HZE particles differing in linear energy transfer." Life Sci Space Res (Amst). 2018 Feb;16:84-92. Epub 2017 Dec 13. https://doi.org/10.1016/j.lssr.2017.12.001 ; PubMed PMID: 29475524 , Feb-2018

Articles in Peer-reviewed Journals	Rabin BM, Polouse S, Bielinski DF, Shukitt-Hale B. "Effects of head-only or whole-body exposure to very low doses of 4He (1000 MeV/n) particles on neuronal function and cognitive performance." Life Sci Space Res (Amst). 2019 Feb;20:85-92. https://doi.org/10.1016/j.lssr.2019.02.001 ; PubMed PMID: 30797437 , Feb-2019